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Citizen forecasting in a mixed electoral system ☆ The 2021 German federal election as a test case

Arndt Leininger [a,*], Andreas E. Murr [b,c], Lukas Stötzer [d], Mark A. Kayser [e]

[a] Chemnitz University of Technology, Germany
[b] Centro de Investigación y Docencia Económicas (CIDE), Mexico City, Mexico
[c] University of Warwick, United Kingdom
[d] University Witten/Herdecke, Germany
[e] Hertie School of Governance, Berlin, Germany

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ABSTRACT

Existing studies show that aggregating citizens' expectations about who will win can predict election outcomes in a majoritarian system. But can so-called citizen forecasting also successfully predict outcomes in mixed-member systems, where constituency results are less important? The existing evidence is mixed and limited in scope. We conducted, therefore, a citizen forecast of the 2021 German federal election by administering an original survey asking citizens who they thought would win in their constituency, what share of the vote each candidate would win in their constituency, and what share of the vote each party would win nationally. Citizens predicted constituency winners and vote shares more accurately than several benchmarks. However, our citizen forecast was based on a non-representative sample from an online-access panel. We conclude that citizen forecasting provides a simple and inexpensive way to predict the various relevant outcomes in mixed-member elections.

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1. Introduction

There are very few surveys for constituency-level election outcomes. Even in majoritarian systems such as France, the United Kingdom, and the United States, most polls are designed to capture only the distribution of vote intentions at the national level. National-level polls, however, cannot easily be used to forecast constituency results because there are, at most, a handful of respondents available for each constituency, while some, if not most, constituencies are not covered at all. Moreover, in mixed electoral systems, such as Germany, these surveys tend to ask respondents only about their vote intentions with respect to the party lists. However, voters in most mixed systems also vote for an individual candidate to represent their district, the constituency vote. Party-list and constituency votes can diverge substantially with strategic voting (Gschwend, 2007), undermining the ability of citizens and politicians alike to predict results. Representative surveys at the constituency level could address this issue but are prohibitively expensive.

We explored a different approach by surveying voter expectations to generate constituency-level predictions of the outcome of the 2021 German federal election. This type of survey is distinct from the more common type that asks respondents about their vote intentions. Our forecast was based on a non-representative sample of citizens in

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* Corresponding author.
E-mail address: arndt.leininger@phil.tu-chemnitz.de (A. Leininger).

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all 299 constituencies. In each constituency, we asked at least 20 people whom they thought would win in their constituency, what share of the constituency vote each party would win in their constituency, and what share of the party-list vote each party would win nationally. We then aggregated their responses, using simple counts to identify modal responses or arithmetic means to arrive at our forecasts. In short, our citizen forecast is based on citizens' expectations about how their fellow citizens will vote rather than their self-reported vote intentions. We published our forecast online three days before the election.[1]

To the best of our knowledge, our forecast is the first-ever application of a vote expectation survey to election forecasting of constituency results in a mixed-member proportional system. In addition, we forecast the distribution of vote shares in each constituency and nationwide. Similar methods have been successfully used for election forecasting in other countries with single-member district electoral systems, such as the United Kingdom (e.g., Murr, 2016) or the United States (e.g., Graefe, 2014; Murr, 2015). We seek to contribute to this literature by addressing the following questions: How well can citizens predict elections in mixed-member proportional electoral systems? How well can they predict vote shares relative to majoritarian outcomes? Moreover, how do their forecasts compare to other approaches to election forecasting? Does the composition of the sample affect the accuracy of the forecasts?

Germany represents a hard case for citizen forecasting. Granted, citizens know their local area and can assess how the constituency has developed demographically and politically over the past four years. For example, even the most ardent supporter of the FDP, a small liberal party that rarely collects more than 10% of the vote, can realistically assess that their party has no chance of winning any constituency seat. However, votes cast in the constituency election do not matter for the distribution of seats of a party, which is the most relevant result. German citizens are less likely to think about the chances of constituency candidates winning the mandate than citizens in majoritarian systems. If citizen forecasting works in Germany, it should also work in other cases where constituency races are more important and voters know more about them.

When forecasting who wins a constituency, we find that citizens do better than simply predicting that the previous winner wins again, although worse than a "proportional swing" approach represented by Zweitstimme.org (Munzert et al., 2017). However, when forecasting a constituency election of particular prominence in the campaign, only citizens predicted correctly. This finding suggests that citizen forecasting can be very useful for forecasting constituency results in mixed-member systems, especially when identifying constituencies that will buck national trends.[2] When forecasting the constituency vote share, citizens forecasted the overall distribution correctly, with a margin of error that we consider low, as there were only about 20 respondents in each constituency. When forecasting the party list vote share, citizens got the distribution right and forecasted with low error, though comparable with vote intention polls conducted shortly before the election. Finally, we find little evidence that the group's composition, whose expectations we aggregate, affects the accuracy of the forecasts, thus further justifying our use of an inexpensive, non-representative online sample.

In summary, we contribute to the growing literature on citizen forecasting by, first, applying the method to a mixed electoral system for the first time; second, forecasting both binary and continuous outcomes; and third, demonstrating the limited relevance of group characteristics for the 'wisdom of crowds'. As such, our findings are relevant beyond the field of election forecasting. Wisdom of crowds approaches are increasingly popular in various disciplines, such as psychology (Keck & Tang, 2020) and economics (Joseph et al., 2021), for making forecasts and studying human reasoning and expectation formation. While election forecasting remains the most prominent application, researchers have also used ordinary citizens' expectations to predict outcomes in other areas, such as economic indicators (Neto & Fontanari, 2023) and sporting events (Brown & Reade, 2019; Peeters, 2018). Citizen forecasting offers a simple and cost-effective way to predict various relevant electoral outcomes in mixed-member systems. We hope our results inspire researchers to apply citizen forecasting to further electoral systems and other significant non-political outcomes.

2. Citizen forecasting: principles and practice

Citizen forecasting promises to be a relatively straightforward approach to forecasting constituency results, but whether it can be a valuable addition to the forecaster's toolkit hinges on the answers to several questions: How much do citizens know about politics? Can they, for instance, predict who will win the election or even what vote share a party will win? If they could, we would have found evidence that citizens know something valuable and challenging to learn about politics. This consideration implies that studying citizen forecasting helps us gauge the public's political knowledge. Why do we think asking citizens for their expectations of the election outcome works?

One reason is that aggregating responses leads to a more precise signal relative to the noise of individual responses, the so-called 'miracle of aggregation'. The basic

[1] The post "Citizens' Forecast for the 2021 German National Election", available at https://aleininger.eu/citizens_forecast2021/, published on 23 September 2021, describes our methods and presented the forecasts. The German newspaper Freie Presse reported on our forecast on the day before the election, see https://www.freiepresse.de/nachrichten/deutschland/bundestags-direktmandate-im-blick-der-demoskopen-das-sagen-die-prognosen-artikel11733130 (last retrieved 20 August 2024). We also published a description of our forecast and our assessment of the election outcome in light of our forecast in a German book chapter (Leininger et al., 2024).

[2] That is, constituencies that a unit or proportional swing approach would forecast incorrectly.

idea is that the law of large numbers also applies when asking a set of citizens to forecast the election, irrespective of whether the forecasting task is discrete ('Who will win the election?) or continuous ('What vote shares will the different parties obtain?').

When asking citizens to predict a discrete event, the underlying theory is Condorcet's jury theorem and its extensions (Murr, 2011). Condorcet's theorem stipulates that if citizens have an equal, larger than 50 percent chance of picking the correct of two alternatives, then the probability that the group makes the correct decision via majority rule approaches unity as the sample size increases. The corresponding aggregation effect also emerges in small groups already: for instance, a group of 25, where each member has a probability of 60% of making the right decision, has a probability of 80% for reaching the correct decision by majority rule. Since Condorcet's original proposition, many refinements of his theorem have been put forward to deal with more complex but realistic situations such as heterogeneous groups or non-binary decisions (Nurmi, 2002, ch. 4). For example, researchers have shown that the theorem holds even when individuals have different probabilities of making the correct decision (Boland, 1989), when individuals' decisions are correlated (Becker et al., 2017), or when individuals have to choose between more than two options (List & Goodin, 2001).

The modern study of citizen forecasting originated with Lewis-Beck and Skalaban (1989), who looked at what proportion of US citizens could correctly forecast the winner of eight US presidential elections. They found that averaging across these elections, 69 percent of respondents to the American National Election Study (ANES) correctly forecasted the national election winner. However, the majority of citizens correctly forecasted six out of the eight elections (75 percent)—an aggregation effect of + 6 points. Murr (2015) then looked at what proportion of US citizens correctly forecasted which presidential candidate would win in their state. He found that 69 percent of respondents to the ANES correctly predicted the winner in their state. A majority of citizens correctly forecasted 82 percent of the states—an aggregation effect of + 13 points. This forecasting performance is impressive because, on average, the ANES survey interviewed only about 30 respondents in each state.

Since this pioneering work, similar findings have emerged in different political systems such as Canada (Temporao et al., 2019), France (Dufresne et al., 2022), Germany (Murr & Lewis-Beck, 2022), Great Britain Lewis-Beck and Stegmaier (e.g., 2011), and Mexico (Murr, 2025). For instance, Murr (2011, 2016) found that most British citizens correctly forecasted which party would win the general election in their constituency. He then predicted that the party, which most respondents saw as the likely winner, would win the constituency and found that most constituencies correctly predicted this way from 1964 to 2015. For instance, he found that 61 percent of respondents correctly predicted the winner in their constituency. In contrast, when aggregating respondents' answers within the constituencies, he correctly forecasted 85 percent of constituencies—an aggregation effect of + 24 points. This forecasting performance is impressive because the underlying survey only interviewed about 28 respondents in each constituency. Overall, several studies have uncovered evidence that a majority of citizens correctly predict the election winner most of the time, at both the constituency and the national level.

Encouraging findings such as those detailed above have spurred comparisons with other approaches to election forecasting. How do citizen forecasts compare to betting markets, vote intention polls, experts, and structural models? Looking at several US presidential elections, Graefe (2014) found that citizen forecasting was among the most accurate forecasting approaches. Looking at several British general elections, Murr et al. (2021) found that citizen forecasts predicted more accurately than commonly used forecasting models based on vote intention polls. This said, looking at Mexican presidential elections from 2000 to 2024 after the transition to democracy, Murr (2025) found that while citizens got better at forecasting over time, they are still not better than vote intention polls. Hence, citizen forecasts of who wins rank among the relatively most accurate approaches to election forecasting, at least in mature democracies.

These findings make us optimistic about applying the citizen forecasting approach to the constituency vote in mixed systems. However, one question remains: Can citizens also forecast vote shares, which is essential in proportional electoral systems? Here, one asks citizens to forecast the vote shares that a party or candidate will win. This task is considerably more complex than simply picking the winner. The reason is that it requires citizens also to get the winning margin right. Nevertheless, the law of large numbers should still apply. If the forecasts are centered on the actual outcome but made with random error, then the mean forecast converges to the actual outcome as the sample size increases.

The evidence of their relative accuracy is mixed for continuous citizen forecasts, though far from complete. For instance, using data from the 2010 British general election, Boon (2012) found that a vote expectation poll predicted national vote shares better than the most accurate vote intention poll. However, using data from the 2015 British general election, Murr (2017) found the opposite: the vote share expectation poll predicted worse than any vote intention polls collected simultaneously. Mixed evidence also emerged in Germany, though the evidence comes from only two elections. Using data from the 2014 German federal election, Ganser and Riordan (2015) found that an expectation poll for the national vote share was less accurate than a vote intention poll. While Graefe (2016) found the same, he also found that vote share expectation polls were more accurate than betting markets (though less accurate than model-based approaches). However, using data from the 2017 German federal election, Lehrer et al. (2019) found that a vote share expectation poll was more accurate in predicting the *Alternative für Deutschland* than a vote intention poll and alternative indirect questioning techniques.

The debate about whether citizens can forecast vote shares has yet to be settled. As data on vote expectations for previous elections are scarce, we applied an

ex-ante citizen forecast to the 2021 German federal election. Because all studies to date have asked citizens to forecast only a single outcome, we decided to focus on the full range of relevant election outcomes. Our survey collected respondents' expectations of who would win their constituency, the vote shares of each candidate in their constituency, and the vote shares of the major parties at the national level. This allows us to conduct a complete test of citizens' predictive abilities in a mixed electoral system.

Another point of contention is the diversity of groups. Theoretically, all that is needed is a modestly large and diverse group of people so that group members' individual biases cancel out in the aggregate. Nevertheless, researchers have wondered whether more educated or knowledgeable groups may provide more accurate expectations (Mellers et al., 2015). Empirical results, so far, are mixed. For instance, Murr (2015) found that weighting individuals' expectations according to their level of expertise increased the accuracy of this citizen forecast for state outcomes in US presidential elections. Similarly, Mongrain (2021) found weighting by competence to improve citizen forecasts for elections in Canada, Germany, Great Britain, and the US. The accuracy of citizens' expectations can also be improved by providing additional information such as poll marginals (Stoetzer et al., 2024) or elite cues (Morisi & Leeper, 2024). In contrast, Ganser and Riordan (2015) did not find that more educated subgroups of their sample generated more accurate predictions.

3. A citizen forecast of the 2021 German federal election

3.1. The case

How many constituencies will each party win in the election? This is the decisive question in majoritarian electoral systems but is also highly relevant to mixed systems. Germany employs a mixed-member proportional (MMP) electoral system for its federal elections, combining majoritarian and proportional representation elements. Voters cast two votes: the so-called *Erststimme* for a constituency candidate and the so-called *Zweitstimme* for a party list.[3] The overall distribution of seats is determined by the part-list vote shares. However, about half of these seats are filled by candidates who win a plurality in one of Germany's 299 single-member constituencies. In contrast, the remaining seats are allocated to parties based on their share of second votes, ensuring overall proportionality. The 2021 national election saw seven viable parties[4] compete for seats in the Bundestag by fielding constituency candidates and party lists. These were the populist radical left Die Linke, the Greens, the social democratic SPD, the Christian-Democratic CDU, its Bavarian sister party CSU, the liberal FDP, and the populist radical right AfD.

Constituency results were also of considerable public interest in the 2021 German federal election run-up despite its mixed-member proportional electoral system. More constituency races than ever before were expected to be very close. Many constituencies were also prominently contested: for instance, in constituency 61 *Potsdam*, Chancellor candidates Olaf Scholz (SPD) and Annalena Baerbock (Greens) competed against each other for the so-called 'direct mandate'. Moreover, the minds of several parties focused on the constituency outcomes: for instance, the Greens were hoping to obtain multiple direct mandates for the first time, and the AfD was hoping to retain many of the constituency seats it won in 2017. Furthermore, the distribution of constituency results would have a decisive influence on the size of the 20th German Bundestag. Would the new Bundestag be even larger than after the 2017 election, which produced Germany's largest-ever national parliament?[5]

The public and pundits alike showed great interest in forecasts of the outcome of the 2021 German federal election. As usual, people were strongly interested in national and constituency-level forecasts for the above reasons. Of course, just as in any other healthy democracy, there is a tremendous public interest in the likely outcome of an election as the election campaign unfolds. Recently, an increasing number of academic and non-academic forecasts in Germany have met this demand for forecasts. Almost all of these approaches focused exclusively on the second vote (*Zweitstimme* in German) at the national level (for an exception, see Munzert, 2017; Neunhoeffer et al., 2020). This focus on the second vote also remains the case for vote intention surveys, which are frequently, although incorrectly, treated as forecasts by the media and interested public.[6]

3.2. The survey

We commissioned online-panel provider *respondi* to survey a non-representative sample of eligible voters[7] about their expectations for the 2021 German federal election. The survey field time was from 6 to 22 September 2021, from three weeks to only four days before the election date on 26 September. We interviewed at least 20 respondents per constituency; the average number of respondents per constituency was 21.6. This number of respondents would be too small to forecast constituency results based on individual vote intentions. However, if one asks about the expected outcome of the election, extant research from the US and the UK, among others,

[3] These German terms translate to first and second votes, reflecting the order in which they appear on the ballot paper.

[4] Germany has an electoral threshold: only parties that win at least 5% of the party-list vote nationally or three constituency seats gain parliamentary representation.

[5] Germany's mixed-member proportional representation system requires that seat shares in parliament are proportional to vote shares at the national and state levels. However, when a party wins many constituency seats, its seat share can surpass its vote share to create so-called *Überhangsmandate* (surplus seats). These are compensated by giving additional seats to other parties until proportionality is reached. Going into the 2021 election, Germany's parliament had 709 seats. After the election, that number rose to 736, more than the 599 seats it would have had there been no *Überhangsmandate*.

[6] To discuss why vote intention polls are not forecasts, see Leininger (2015).

[7] That is, German citizens aged 18 or older.

Table 1
Number of constituency seats predicted by our citizen forecast versus the actual number of constituency seats won by parties and the average vote share for parties' constituency candidates as predicted by respondents versus actual vote shares received.

Party	Seats		Vote shares	
	Forecast	Result	Forecast	Result
CDU/CSU	174	143	26.5	28.6
SPD	98	121	24.0	26.4
Too close to call	13	0	0.0	0.0
Greens	7	16	14.9	14.0
AfD	5	16	9.4	10.1
Linke	2	3	7.7	5.0
FDP	0	0	10.2	8.7
Other	0	0	7.3	7.3

has shown that even small numbers of respondents per constituency can provide reliable forecasts (Murr, 2015, 2016). Our questionnaire contained only a few items and took respondents at most five minutes to complete.

3.3. Which candidates will win?

We begin with our forecast of the winners of the constituency races. The question about the direct mandate winner was as follows: "Which candidate will win the most votes in your constituency in the Bundestag election?" Respondents could choose from a list of the candidates in their constituency to answer the question. We listed only candidates from the major parties, ignoring independent candidates and candidates of smaller parties as none of them had won a direct mandate in recent decades. The modal response in a constituency became our forecast of the winner of the direct mandate.

In most constituencies (286 out of 299), we were able to make a forecast, as the plurality of respondents' expectations favored one single candidate. Two candidates were tied in respondents' expectations in a few remaining constituencies (13 out of 299). We labeled these constituencies as "too close to call". Such tied expectations among respondents within a constituency foreshadow electoral outcomes to some extent, as the vote margin between the winner and runner-up in these thirteen constituencies was significantly lower – on average, 3.2 percentage points, compared to an average of 10 percentage points in all other constituencies. The second and third columns of Table 1 show that we overestimated the number of seats to be won by the CDU/CSU and underestimated the number of seats the other major parties won.[8] Nevertheless, our forecast captured the relative distribution of direct mandates: as predicted, the CDU/CSU won most of them, followed by the SPD, and then the Greens, AfD, and the Left, though we did not predict the tie between the Greens and the AfD. We also correctly predicted that neither the FDP nor any other small party would win seats.

Overall, in 74.9% (224 out of 299) of the constituencies, the candidate forecasted as the winner by a majority of respondents in that constituency did win the constituency.

[8] Figure B.1, in the online appendix, presents our forecast for the constituency winner based on respondents' expectations (left panel) and the actual election outcome (right panel).

Table 2
Accuracy of individual and group forecasts of which candidate will win the direct mandate: % of correct predictions by individuals or majority of respondents in a constituency (group).

	Individual	Constituency
Correct	46.8	74.9
Incorrect	53.2	20.7
Tied	–	4.3

But what proportion of individuals correctly predicted the winner? Are groups better at predicting than individuals? Table 2 shows the proportion of respondents who correctly or incorrectly forecasted the outcome in their constituency compared to our forecasts at the constituency level. Only slightly less than half of our respondents correctly forecasted the winner in their constituency. When aggregating to the modal response at the constituency level, we can correctly forecast 75 percent of all constituency results – an aggregation effect of almost + 30 points.

3.4. What percentage of votes will each candidate receive?

For our forecast of the candidates' vote shares in the constituency, we asked respondents: "What percentage of first-past-the-post votes will the various candidates receive in your constituency?" To answer this question, respondents could distribute vote shares among the candidates of all parties represented in the Bundestag and a residual category "Others". All percentages provided had to add up to 100% before respondents could submit their answers.[9]

We first examine national averages to provide insight into how accurately citizens can form expectations about vote shares at the constituency level. In other words, we pool all respondents and compute the mean of each party's expected vote share. We then pool all party candidates and compute the mean of their actual vote share. The final two columns of Table 1 show that the national average of our respondents' expectations was remarkably close to the actual outcome.

With this encouraging result in mind, how well did citizens predict the vote share in their constituency? Were groups better than individuals in predicting the vote share? To answer these questions, we calculate the forecasting error between the actual election results and the respondents' expectations about the vote share for the parties' constituency candidates. More specifically, we calculate the Root Mean Square Error (RMSE) between the vote share predicted by our respondents and the actual result.[10]

The second and fourth columns of Table 3 show that individual respondents forecast with a relatively large error. For instance, when forecasting the vote share of

[9] We could have also used this information in our forecast of who wins the constituency by, for instance, breaking ties between candidates with the same number of forecasts. We did not do so at the time to keep our forecasting procedure as simple as possible.

[10] In Table B.1 in the appendix, we present results using the Mean Absolute Error (MAE), leading to the same conclusions discussed here.

Table 3
Root Mean Squared Error of individual and group forecasts of a candidate's constituency vote share (first vote) and the party list vote (second vote).

	Constituency vote		Party-list vote	
	Individual	Constituency	Individual	National
SPD	12.6	6.2	10.4	0.7
CDU/CSU	13.2	5.1	10.5	0.7
Greens	10.0	4.9	8.3	1.6
AfD	11.3	3.7	10.8	0.2
FDP	7.6	2.6	6.9	1.1
Linke	7.5	3.7	7.6	2.9
Other	13.3	4.5	11.0	2.3
Overall	11.0	4.5	9.5	1.6

candidates from the CDU/CSU, individual respondents' average deviation from the actual vote share won by that candidate as measured by the RMSE was 13.2. While the RMSE was less for smaller parties, such as the *Die Linke* (about 7.5), this is still a significant error. Overall, the RMSE across all parties is about 11.

These results, however, only focus on mistakes made by single voters. Yet, the key idea behind citizen forecasting is that, on average, citizen forecasting can be very accurate because erroneous individual predictions, which over- or underestimate a candidate's or party's performance, tend to cancel each other out. We expect groups of respondents to be more accurate on average than we expect single respondents to be on average. Indeed, we obtain much more accurate forecasts when we aggregate individual respondents' predictions at the constituency level. The third and fifth columns of Table 3 show the results. Looking at the average vote shares predicted for constituency candidates by all respondents in a constituency, in the third column, the RMSEs are considerably lower. Looking again at the CDU/CSU, as we move from individual to constituency forecasts, the RMSE drops from 13.2 to 5.1 – a reduction in error by more than a half. Table 3 demonstrates that the improvement in accuracy through aggregation accrues to all parties. Moving from the individual to the group forecast reduces the RMSE from 11 to 4.5, a reduction by more than half. This improvement in accuracy is a remarkable result, as our forecasts are based on only about 20 respondents per constituency.

Finally, we look further into how well our citizen forecast anticipated how close the constituency races were. In Fig. 1, we plot the actual margin between first- and second-place candidates against the margin between first- and second-placed candidates in our citizen forecast. This visualization affords us with a few more insights into our forecasting methodology. First of all, perhaps not surprisingly, the majority of respondents in a constituency were less likely to identify the winner in constituencies that they expected to be close and that ended up being close. As can be seen in the figure, most incorrect forecasts cluster in the bottom-left quadrant. Second, respondents generally underestimated the closeness of constituency races. As can be seen in the figure, most dots lie above the dashed line, indicating equality of forecasted and actual vote margin, implying that the margin in most cases ended up being larger than expected. On average, respondents underpredicted the vote margin by 3.87 percentage points. Third, and after considering this under-prediction, citizen forecasting can predict which constituency races will be close as there is a significant, almost one-to-one mapping between forecasted and actual vote margins on average. As the forecasted vote margin increases by one percentage point, the expected actual vote margin increases by one percentage point on average, too.[11]

3.5. What percentage of votes will each party receive nationwide?

Finally, we look at our forecast of the party-list vote. How well did our respondents foresee distributing the more important party-list vote share? To measure expected second vote shares, we asked respondents: "What percentage of the vote do you expect the various parties will receive nationwide?" Again, respondents could distribute vote shares to the different parties in steps of one percentage point, while the survey software enforced that these had to sum up 100% before respondents could submit their answers. As a forecast of the national election outcome, we calculated the unweighted arithmetic mean of all responses per party.[12]

We conclude this initial evaluation of our citizen forecast by calculating the forecasting error between the actual election results and the respondents' expectations about the party-list vote share (second vote). As for the first vote, we calculate the Root Mean Squared Error (RMSE) between the vote share predicted by our respondents and the actual result. Again, we do so separately for individual and group predictions (the group prediction is simply the arithmetic mean of all individual predictions). In Table 3, we see again an improvement in accuracy when aggregating individual responses. The last column indicates the RMSE of the national forecast for the actual result of the party list vote. The entry in the last row shows the overall RMSE, which represents a drastic increase in accuracy over an average respondent.

Table 3 also allows us to compare the accuracy of individual respondents' predictions for the constituency vote share and the party list vote share. Across all parties and for each party individually, citizens find it easier to predict the vote share at the national level than the vote shares for the candidates in their constituency. Three out of four respondents predicted the party list more accurately than the constituency vote shares.[13] This finding is, in general, not surprising. However, given the prominence of vote intention polls and the wide availability of information

[11] Results of a bivariate regression of actual vote margins on forecasted vote margins are as follows: $E(\text{actual vote margin}) = 3.87^{***} + 1.05^{***} \times$ forecasted vote margin; $N = 299$, $R^2 = 0.37$, Significance levels: *** 0.001 ** 0.01 * 0.05.

[12] Figure B.2 shows the distribution of respondents' answers.

[13] For 74.9% of respondents, the RSME of their predictions for the five major parties plus the residual category 'others' was larger for the constituency vote shares than for the party list vote shares. The corresponding share is 72.9% when calculating and comparing the MAEs of individual respondents.

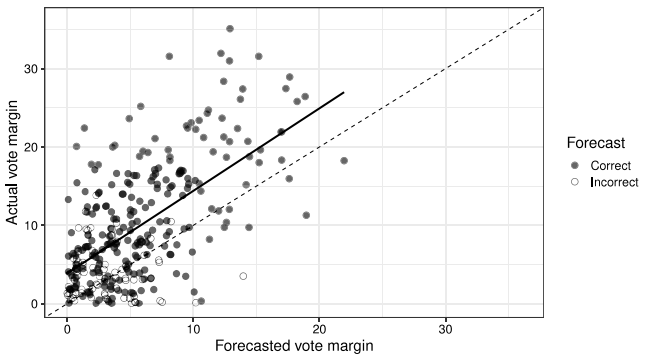


Fig. 1. Vote margins between first- and second-place candidate as forecasted by respondents in a constituency, on the *x*-axis, and in the actual election outcome, on the *y*-axis. The dashed line denotes a perfect one-to-one relationship, while the solid line indicates the linear relationship between forecasted and actual margins.

about the national results of the previous, one might have expected this difference to be even larger.

Having obtained some first insights into the veracity of citizen forecasting in mixed-member proportional systems, we now compare our approach against alternative approaches to forecasting elections in these systems. Such comparisons will allow us to learn about forecasting methodology and the specifics of the election under study.

4. Evaluating the citizen forecast

4.1. Forecasting the constituency winner

In contrast to the nationwide second vote results, forecasting models for direct mandates have appeared only recently. To our knowledge, ours is only the second scientific attempt to forecast constituency results in Germany. With Zweistimme.org, only one other scientific forecasting model for the federal election forecasted constituency results using an entirely different approach.[14] Gschwend and co-authors (2022) use a proportional swing model to translate their forecast for the national party-list vote into forecasts of individual constituency results. The authors have described their method for constituency forecasting in various scientific publications (Munzert et al., 2017; Stötzer et al., 2019). They also publicized their forecasts for the 2021 election on the website of the German newspaper *Süddeutsche Zeitung*. They began providing their first forecast a few weeks before the election and updated it continuously as new polling results were released.

To allow for a fair comparison, we draw on their forecast from Friday, 24 September 2021, two days before the election and one day after our forecast was published. In addition to these two scientific forecasting models, a straightforward approach to forecasting predicts that the same party that won a constituency in 2017 will win it again in 2021. This simple approach provides a benchmark that every scientific forecasting model should aspire to surpass. How do these three forecasts compare against each other in predicting who wins the constituency?

Table 4
Correct predictions of constituency winners (first vote) made by citizen forecasting, Zweitstimme.org, and the simple benchmark (predicting the previous winner to win again) in all constituencies (second column) and constituencies that saw a change in the winning party from 2017 to 2021 (third column).

Approach	Correct predictions	
	All	Change
Zweitstimme.org	235 (78.6%)	53 (54.6%)
Citizen Forecast	224 (74.9%)	39 (40.2%)
Simple benchmark (past winner)	202 (67.6%)	–

Table 4 shows the number and proportion of correctly forecasted constituencies by our citizen forecast, Zweitstimme.org, and the simple benchmark. The simple benchmark correctly predicted about 68 percent of constituencies. Citizen forecasting correctly predicted 75 percent of constituencies. This result implies that citizens do not forecast the previous winner to win again; they can perceive change and stability. Zweitstimme.org further improves upon citizen forecasting by correctly predicting 79 percent of constituencies. Again, this implies that Zweitstimme.org predicted some changes correctly. Overall, the results suggest that citizen forecasting improves upon simply predicting the previous winner to win again. Nevertheless, to reach the accuracy of Zweitstimme.org, citizen forecasting would need to be improved, either by interviewing more citizens or in some other way, such as weighting individual respondents (Murr, 2015).

As indicated above, a good forecasting model correctly forecasts changes in the winning party. In the 2021 federal election, about one-third of constituencies (97 out of 299) changed hands from one party to the next compared to 2017. The simple benchmark (predicting the previous winner to win again) cannot predict change. Both citizen forecasting and Zweitstimme.org predicted changes. How good was their prediction? The last column

[14] In addition to the scientific approaches discussed here, two other private or commercial providers offered constituency forecasts ahead of the election. election.de (www.election.de) publishes, among other things, constituency forecasts, and one has to pay to view detailed results. Wahlkreisprognose (www.wahlkreisprognose.de/), similarly, offers constituency forecasts and other services for a fee. According to their website, they use a prediction model that considers demographic trends, survey data, and unspecified further information. Because neither of them discloses their methodology, we do not incorporate them in the comparisons in this section.

of Table 4 shows the forecasts from our citizen forecasting model and Zweitstimme.org for the 97 constituencies that changed hands. Zweitstimme.org's proportional swing approach does slightly better in predicting change than citizens. However, this may also be due to the unusually large shift in first and second votes from SPD to CDU in 2021. We should note that both approaches underpredicted the SPD's success.

We continue to focus on change and further illustrate the difference between both approaches by looking at one high-profile race for a direct mandate spotlighted by the media. In the spotlight was constituency 196 *Schmalkalden-Meiningen - Hildburghausen - Sonneberg* because it featured two attention-catching candidates: Hans-Georg Maaßen and Frank Ullrich. Maaßen is the former chief of the federal office for the protection of the constitution, who recently turned right-wing populist and competed as the constituency candidate for the local CDU against the expressed wishes of the national CDU. Ulrich is a former biathlon athlete turned politician who competed as the candidate for the SPD. Respondents from this constituency correctly predicted that Ullrich would win against Maaßen, which meant that the CDU lost the district to the SPD. While Zweistimme.org correctly anticipated a rise in votes for the SPD, it still predicted a win for the CDU. However, in constituency 196, the SPD more than doubled its vote share in the first vote. At the same time, the CDU lost significantly more votes than elsewhere, most likely due to the controversial candidature of Hans-Georg Maaßen. This observation suggests that the citizen forecasting approach may have an advantage in identifying constituency results that do not follow national trends. While a proportional swing approach may work well when and where national issues dominate the campaign, a citizen forecasting approach, in contrast, may work better when and where local issues or candidates matter more.

4.2. Forecasting the party list vote share

We conclude our evaluation with a comparison of forecasts for the national party-list vote as this outcome, in Germany's mixed-member proportional system, determines the overall seat distribution in parliament. Hence, it gets the most attention in the media and is of the greatest interest to the general public. Because of this, most forecasts focus on this outcome. When focusing on this outcome, we can broaden the range of approaches we compare. Here, we also consider the other forecasting approaches – structural and aggregation models – and a vote intention poll, which many, although wrongly, interpret as forecasts. The vote intentions come from the final pre-election poll of the prominent *Politbarometer* conducted for Germany's public broadcaster ZDF.

Table 5 summarizes the forecasts, with the final rows presenting the RMSE across all parties.[15] Again, Zweitstimme.org (Gschwend et al., 2022) is the most accurate of all forecasting approaches, followed by our citizen forecast and the PollyVote (Graefe, 2022), tied in second place. The final Politbarometer poll, published three days before the election, is more accurate. Jérôme et al.'s (2022) 'Political Economy' model and Kayser, Vlasenko and Leininger's (2022) 'Länder' model fare considerably worse. As does the other citizen forecast by Murr and Lewis-Beck (2021). Their forecast is derived from regression models, which predict a party's vote share with two predictors: the proportion of Politbarometer respondents who say that this party will win and a dummy indicating whether a grand coalition forms the incumbent government. Nevertheless, one must remember that those forecasts were already issued in June of the election year. That is at a time when vote intention polls still had the SPD at less than 20%, putting the CDU in a comfortable lead with well over 25%.

We conclude that citizens' expectations about the national election outcome are reasonably accurate. However, they provide little improvement over a traditional vote intention poll, at least when both are conducted temporally close to the election. It would be interesting to see how both compare when conducted a few weeks instead of days before the election. However, as our results presented in the preceding subsections have shown, citizen forecasting shows its strength in predicting subnational outcomes, which national-level polls cannot do.

5. Group composition and forecasting accuracy

Our citizen forecast was based on an inexpensive, non-representative online panel. Our only requirement for the online access panel provider was to secure 20 respondents per electoral district. Our sample, for instance, is younger and more educated than the population. About 3 percent of our sample are 75 years or older, whereas about 13 percent of the population are – a difference of about 10 percentage points. The sample is also more educated than the population. About 49 percent of our sample have the highest school degree (*Fachhochschul-* or *Hochschulreife*), whereas only about 36 percent in the population do – a difference of about 13 percentage points. An exact binomial test at the 0.001 level rejects both null hypotheses that the proportions are the same in the sample and population.

While our forecasting results are encouraging, one may wonder if we could have achieved even better results by applying demographic quotas or sampling politically interested or knowledgeable respondents. These questions align with a broader debate in the literature about whether the accuracy of citizen forecasts for election outcomes depends on the collective knowledge within the group and the diversity of its members (Ganser & Riordan, 2015; Lehrer et al., 2019; Murr, 2011, 2015; Surowiecki, 2004).

The study of citizen forecasting has found several characteristics of citizens that explain individuals' forecasting accuracy. For instance, in their classic study, Lewis-Beck and Skalaban (1989) find in a multivariate analysis that more educated US citizens are more likely to forecast correctly the winner of the US presidency than less educated citizens, holding their party identification, vote intention, and interview date constant.

[15] See Table B.2 in the appendix for forecasting errors expressed as MAE.

Table 5
Forecasts for the national party-list vote share of our citizen forecast and other approaches compares against the final pre-election Politbarometer poll and the election result.

Party	Citizens' Forecast	Zweitstimme	PollyVote	Political Economy	Länder Forecast	Lewis-Beck and Murr (2022)	Politbarometer	2021 Result
SPD	25.0	26.0	24.1	20.6	19.9	21.0	25.0	25.7
CDU/CSU	23.4	23.0	24.1	32.1	29.9	34.0	23.0	24.1
Greens	16.4	16.0	15.5	17.9	13.6	18.0	16.5	14.8
FDP	10.4	11.0	10.8	7.5	9.2	11.0	11.0	11.5
AfD	10.5	11.0	10.9	–	12.0	–	10.0	10.3
Linke	7.9	6.0	7.2	8.8	8.6	13.0	6.0	4.9
Other	6.4	7.0	5.0	13.1	6.6	3.0	8.5	8.7
RMSE	1.7	1.0	1.8	–	3.7	–	1.0	0.0
RMSE (AfD + Other)	1.8	0.9	1.8	5.3	3.8	8.7	1.0	0.0

Note: Forecasts as of: 23 September for our citizen forecast, 24 September for Zweitstimme.org, 23 September for PollyVote, June for the Political Economy model, the Länder Forecast, and the citizen forecast by Murr and Lewis-Beck (2021). The Politbarometer poll was published on 23 September 2021. Row RMSE provides the mean absolute error calculated over all parties. For the Political Economy Model and the citizen forecast by Murr and Lewis-Beck (2021), the category 'Others' includes the AfD. The final row, therefore, provides the RMSE that results when adding up the vote shares received by the AfD and other parties.

Nevertheless, some recent studies have shifted from explaining why individual citizens forecast accurately to why groups do so (Lehrer et al., 2019; Murr, 2011, 2015). Two reasons justify this shift. First, citizen forecasting predicts elections by grouping forecasts: it aggregates individual citizen forecasts by taking, for instance, individual forecasts' modal or mean values. Second, analyzing the accuracy of group forecasts instead of individual forecasts avoids the 'atomistic fallacy' (Hox, 2002, 4): one cannot infer that the same statistical relationships observed in individuals also hold in groups.

To illustrate the atomistic fallacy, consider education's role in forecasting again. We have ample evidence that more educated citizens forecast with greater accuracy (e.g., Lewis-Beck & Skalaban, 1989). But from this finding, it does not follow that selecting a group of the most educated citizens will forecast with the greatest accuracy. Indeed, we have theoretical reasons and empirical evidence to the contrary. In terms of theory, Murr (2011) argues, based on Surowiecki (2004), that the diversity of groups ensures that they bring different bits of information to bear on the problem. So that when aggregating citizens' forecasts, their errors are canceled out. In terms of evidence for this theory, consider Lehrer et al. (2019) who study citizen forecasts of the vote share of the *Alternative für Deutschland* in 2017. They found that the average forecast of university graduates was further from the actual vote share than the average forecast of the entire sample. Hence, a more educated but less diverse group (university graduates) is forecasted to be worse than a less educated but more diverse group (the full sample).

As a result, to understand why citizen forecasting works and whether we could improve it with different sampling strategies, we need to study citizen forecasting at the group level, that is, the level to which we aggregate the individual citizen forecasts. Only a few studies have done this to date, and none considers the level and diversity of group characteristics simultaneously in a multivariate framework.

Consider the following two typical examples. First, Murr (2011) tests whether the diversity of group characteristics – such as variation in education – predicts group accuracy in multivariate analysis but does not control for their levels – such as the average education. Second, Ganser and Riordan (2015) test whether a subgroup of highly educated citizens forecast better than a subgroup of less educated citizens (and the full sample). Still, they do not control for variables such as age and interview date in a multivariate analysis. Hence, we do not know the relative importance of the level and the diversity of group characteristics for the group's forecasting accuracy. To remedy this situation, we conduct the first multivariate analysis of forecasting accuracy incorporating both the level and diversity of group characteristics.

Our sampling strategy resulted in variation in the size and composition of groups, enabling us to test the association of group characteristics with the accuracy of their forecasts.[16] For the following analysis, we calculated the forecasting error in each of the 299 constituency subgroups separately for their forecast of constituency and party list vote shares. We modeled the forecasting error – calculated as the Root Mean Squared Error (RMSE) – as a function of the average levels and the diversity of the respondent characteristics available to us. For the levels, we considered the number of respondents, the proportion of women, the average age, the proportion of respondents with the highest school degree (*Fachhochschul-* or *Hochschulreife*), the distance of the average left–right placement from the mid-point, the average political interest, and the average number of days between interview dates and election date per constituency. To measure subgroup diversity, we followed Murr (2011) by calculating the entropy for the categorical outcomes of gender categories and school degrees, and the standard deviation of age, political interest, left–right placements, and the number of days between interview dates and election date per constituency.

We then regressed group-level forecasting errors on these group-level measures. Our sample size for this analysis is 299, as this is the number of constituencies for the national parliament in Germany. Table 6 displays the

[16] See Table B.4 in the appendix for summary statistics for all variables included in the regression models presented here.

Table 6
Explaining the forecasting error of groups as measured by the Root Mean Squared Error (RMSE) through linear regression of forecasting errors on group characteristic ($n = 299$)

	Constituency vote			Party list vote		
	Coef.	SE	*p*-value	Coef.	SE	*p*-value
Intercept	9.80	3.36	0.00	4.82	2.01	0.02
Level						
Number of respondents	−0.09	0.08	0.23	−0.12	0.05	0.01
Gender	−1.00	0.83	0.23	0.82	0.50	0.10
Age	−0.10	0.02	0.00	−0.01	0.01	0.41
Education	−2.45	0.70	0.00	−0.98	0.42	0.02
Interest	0.52	0.42	0.21	0.22	0.25	0.37
Ideology	0.19	0.23	0.41	0.41	0.14	0.00
Date	0.02	0.07	0.75	0.01	0.04	0.74
Diversity						
Gender	0.59	1.63	0.72	1.09	0.97	0.26
Age	−0.04	0.04	0.30	−0.04	0.03	0.14
Education	0.34	0.48	0.48	−0.79	0.29	0.01
Interest	−0.37	0.54	0.50	0.29	0.32	0.37
Ideology	0.40	0.26	0.12	0.22	0.15	0.14
Date	0.09	0.10	0.36	−0.01	0.06	0.83
Residual std. dev.	1.33			0.80		
R^2	0.09			0.13		

Note: Level variables include the number of respondents, the proportion of women, the average age, the proportion of respondents with the highest school degree (*Fachhochschul-* or *Hochschulreife*), the distance of the average left–right placement from the mid-point of the scale, the average political interest, and the average number of days between interview dates and election date per constituency. *Diversity* variables include the entropy of gender categories and school degrees, and the standard deviation of age, political interest, left–right ideology, and the number of days between interview dates and election date per constituency.

regression results. As can be seen in Table 6, four group characteristics reach statistical significance at the 5 percent level: the number of respondents, the average age, the distance of the average ideological self-placement from the mid-point of the scale, the proportion with the highest school degree, and the diversity in school degrees. However, the results vary between the constituency and party list votes.[17]

We turn first to the constituency vote. Groups with a higher average age forecast better – with a lower RMSE – than groups with a lower average age. Groups with a higher proportion of respondents with the highest school degree forecast better than groups with a lower proportion. Larger groups forecast the party list vote better than smaller groups. Groups whose average ideology is closer to the mid-point forecast better than groups whose average ideology is farther away. Also, groups with a higher proportion of respondents with the highest school degree forecast better than groups with a lower proportion. However, groups with a more diverse educational background are forecasted better than groups with a less diverse educational background.

To compare the relative importance of these predictors, we compute predictive differences. The predictive difference is the expected value of the outcome at the observed maximum value of the predictor of interest minus the expected value of the outcome at the observed minimum value of this predictor, holding the values of the other predictors constant.

Table 7 displays the point estimates of the predictive differences for each predictor, together with 95 percent confidence intervals. Turning first to the constituency vote and focusing on statistically significant predictors, Table 7 shows that the predictive difference for the average age is about −2 and for the proportion with the highest school degree about −1.8 points (plus or minus about 1.0). To illustrate the meaning of these predictive differences, consider the one for education. Here, we compare two otherwise identical groups, except that one has the maximum observed proportion of respondents with the highest school degree in the data, and the other has the minimum. We estimate that the more highly educated group predicts the constituency vote with an RMSE of 1.8 points lower than that of the less educated group.

Our sample is younger and more educated than the population, yet it forecasts accurately. The predictive differences for the average age and the average education explain why. They show how the loss in forecasting accuracy due to a decrease in the average age can be (partially) offset by a gain in forecasting accuracy due to an increase in the average education.

To see this (partial) offset, note that constituencies with a lower average age are associated with a higher level of education. We estimate that the expected average age equals $54.92 - 11.73$ times the proportion with the highest school degree, with standard errors of 0.80 and 1.59, respectively. Hence, comparing two constituencies that differ by 10 percentage points in proportion to the highest school degree, the more educated constituency has, on average, an average age that is about 1 year lower than the less educated one. As a result, the loss in accuracy due to lower average age is, in practice, partially offset by a gain in accuracy due to higher levels of education.

To illustrate, consider again the increase in the proportion with the highest school degree from its minimum (0.2) to its maximum (0.9). Plugging this increase of 0.7

[17] In Table B.3 in the appendix, we present results using the Mean Absolute Error (MAE), leading to the same conclusions discussed here.

Table 7
Summaries of the predictive difference for a group's Root Mean Squared Error (RMSE).

	Constituency vote		Party list vote	
	Estimate	95% Interval	Estimate	95% Interval
Level				
Number of respondents (20 → 26)	−0.56	[−1.47, 0.35]	−0.73	[−1.27, −0.18]
Gender (0.2 → 0.8)	−0.58	[−1.53, 0.36]	0.48	[−0.09, 1.04]
Age (38.5 → 58.5)	−1.97	[−2.91, −1.03]	−0.24	[−0.80, 0.32]
Education (0.2 → 0.9)	−1.76	[−2.74, −0.78]	−0.70	[−1.29, −0.12]
Interest (3.0 → 4.2)	0.64	[−0.36, 1.65]	0.27	[−0.33, 0.87]
Ideology (0.0 → 1.7)	0.34	[−0.46, 1.13]	0.72	[0.24, 1.19]
Date (8.9 → 18.8)	0.22	[−1.14, 1.58]	0.14	[−0.68, 0.95]
Diversity				
Gender (0.5 → 0.9)	0.21	[−0.92, 1.34]	0.39	[−0.29, 1.07]
Age (8.5 → 20.9)	−0.56	[−1.61, 0.49]	−0.48	[−1.11, 0.15]
Education (0.5 → 1.6)	0.36	[−0.65, 1.38]	−0.86	[−1.47, −0.25]
Interest (0.6 → 1.4)	−0.30	[−1.15, 0.56]	0.24	[−0.28, 0.75]
Ideology (1.0 → 2.8)	0.73	[−0.17, 1.62]	0.40	[−0.14, 0.94]
Date (0.5 → 6.3)	0.54	[−0.62, 1.70]	−0.07	[−0.77, 0.62]

Note: Values in columns labeled 'Estimate' represent the change in expected forecasting error when moving from observed minimum to maximum value on the predictor reported in the left-most column while holding all the other predictors constant. For variable definitions, see Table 6.

into the estimated regression equation reported in the previous paragraph implies an expected decrease in the average age of about 8 years. Now, if both the average age and the proportion with the highest school degree change like this at the same time, then, according to the estimates underlying Table 7, the expected reduction of forecasting error for the constituency vote nets only about 0.93 with a standard error of about 0.47 – about half the value if age remained constant (cf. Table 7). This finding supports the view that to understand why citizen forecasting works and how to improve it with sampling, we need to study the role of group characteristics, considering several of them simultaneously.

Moving on to the party list vote and focusing again on statistically significant predictors, Table 7 shows that the predictive difference for both the number of respondents and the proportion with the highest school degree is about −0.7, for the distance of the average ideology to the mid-point is about +0.7, and for the diversity of school degrees is about −1.0 points (plus or minus about 0.5). The two predictive differences for education show again how the gain in forecasting accuracy due to an increase in the proportion with the highest school degree can be offset by a loss in forecasting accuracy due to a decrease in the diversity of school degrees.

To see this offset, note that constituencies with higher proportions with the highest school degrees are associated with less diverse school degrees. We estimate that the expected diversity of school degrees equals 1.25 − 0.30 times the proportion with the highest school degree, with standard errors of 0.04 and 0.07, respectively. Hence, comparing two constituencies that differ by ten percentage points in proportion to the highest school degree, the more educated constituency is, on average, 0.03 points less educationally diverse than the less educated one. As a result, the gain in accuracy due to higher levels of education is, in practice, offset by a loss in accuracy due to lower diversity in education.

To illustrate, consider again the increase in the proportion with the highest school degree from its minimum (0.2) to its maximum (0.9). Plugging this increase of 0.7 into the estimated regression equation of the previous paragraph implies an expected decrease in education diversity of about 0.2. Now, if both the level and the diversity of education change like this at the same time, then, according to the estimates underlying Table 7, the expected reduction of forecasting error for the party list vote nets only about 0.53 with a standard error of about 0.30 – a statistically insignificant value. This finding reinforces the view that to understand why citizen forecasting works and how to improve it with sampling, we need to study the role of group characteristics, considering their level and diversity simultaneously.

Overall, most group characteristics did not matter. Moreover, two studies that did- education and age – suggest that making the groups more unrepresentative of the population – more educated and older – results in better forecasts, as long as one ensures sufficient diversity in educational backgrounds. In addition, we also found how unrepresentative characteristics can pull the accuracy in different directions and ultimately offset each other. Hence, these results further support the view that citizen forecasting based on non-representative online samples can be inexpensive and reliable for predicting election outcomes in constituencies.

6. Conclusion

Researchers have successfully applied citizen forecasts to predict election outcomes in majoritarian systems. We sought to contribute to and extend this literature by, for the first time, conducting a citizens forecast in a mixed electoral system to address the following questions: How well can citizens predict elections in mixed-member proportional electoral systems? How well can they predict vote shares relative to majoritarian outcomes? Moreover, how do their forecasts compare to other approaches to election forecasting? Does the composition of the sample affect the accuracy of the forecasts?

Our citizen forecast is rooted in citizens' expectations about their fellow citizens' voting behavior. Instead of asking respondents whom they would vote for, we asked them who they thought would win in their constituency, what vote share a candidate would win in their constituency, and what vote share a party would win nationally. We collected the responses to these questions in the weeks before the German federal election in September 2021. We then used these responses to forecast the winner in Germany's 299 constituencies and the constituency and party list vote shares. We released our forecast to the public three days before the election.

How did citizens do? In forecasting who would win the constituency, citizens did better than a simple prediction that the previous winner would win again but worse than the proportional swing model of Zweitstimme.org. When forecasting vote shares in the constituency, citizens did remarkably well, given that there were only about twenty respondents per constituency. As far as we know, this is the first time researchers asked citizens to predict vote shares at the constituency level. Finally, citizens forecasted with a relatively low error in forecasting vote shares nationwide and compared favorably to some professional forecasting models. However, some of these models forecasted with more lead time. Overall, our citizen forecast was a simple and inexpensive but accurate way to forecast the various election outcomes in a complex electoral system such as Germany's.

These findings contribute to our understanding of citizen forecasting in two ways. First, prior studies conducted in majoritarian systems such as Great Britain or the United States found that most citizens can forecast the winner (e.g., Graefe, 2014; Murr, 2011, 2016). However, it needed to be clarified whether citizens could do the same in mixed-member proportional systems, where the constituency winner is less important politically. Even in this hard case, we showed that a relatively small sample of respondents correctly forecasted who would win. Second, prior studies focusing on whether citizens could forecast vote shares yielded mixed results (e.g., Ganser & Riordan, 2015; Lehrer et al., 2019). Our results tilt the scales in favor of citizens, if only slightly. We showed that citizens could accurately forecast vote shares in their constituency and nationwide. We also demonstrated that national citizens' forecasts compared favorably with alternative forecasting approaches. Nevertheless, some comparisons with other approaches still need to be completed: some approaches forecasted less well but had more lead time than ours. More research is necessary to compare these approaches at different lead times.

Our results also allow us to draw practical implications for implementing citizen forecasting. In particular, we find little evidence that the composition of the group whose expectations we aggregate affects the accuracy of forecasts. We find only four group characteristics that predict forecasting error, depending on whether constituency or party list votes are forecasted: larger, more educated, more diversely educated, and older groups predict with less error than smaller, less educated, more homogeneously educated, and younger groups. We also found that some unrepresentative characteristics pulled the forecasting accuracy in an oppositive direction in our unrepresentative sample, canceling each other out. These results, we believe, further justify our use of an inexpensive, non-representative online sample without quotas. Thus, generating good forecasts from convenience samples collected from services such as MTurk may even be possible.

Declaration of Generative AI and AI-assisted technologies in the writing process

While preparing this work, the authors used Grammarly and DeepL to improve the manuscript's language. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2025.03.007.

Data and code availability

All data and code necessary for replicating the results presented in this paper are available at Harvard Dataverse: https://doi.org/10.7910/DVN/XKHTYB.

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